INDEX

KEY

page number font	**boldface**	main subject treatment
	italic	figure
suffixes	ff	"and pages following"
	g	graph
	p	photograph
	s	screen shot ('scope or spectrum analyzer)
	t	table

A
abbreviations, 1166
ABEL, *see* programmable logic device (PLD)
absolute-value circuit, 257
abuse, *see* argument
ac power control
  microcontroller, **1062–1065**
ac powered
  dc power supply topologies, *630*
accelerometer, *1078*, 1079p, 1082–1084
acoustic ranging, *207*
acronyms, 1166
active-HIGH, active-LOW, *see* logic
Adafruit, 1082
ADC, *see also* converter, AD/DA, **900–956**, 1061, 1082
  ac monitor, 943–944
  bridge-sensor oriented, 945–946, *947*
  choosing, **938–940**
  comparison, **938–940**
  data-acquisition system (DAQ), **946–955**
    islolated SPI, 950–951
    multiplexed 16-channel SAR, *948*, **946–950**
    parallel 8-channel ΔΣ, *953*, *954*, **952–955**
    parallel 8-channel SAR, *951*, *952*, **950–952**
  delta-sigma, *see* delta-sigma
  design examples, **946–955**
  differential-input, 380
  flash, *903*, **902–907**
    driving (design example), *906*, 904–907
  folding, 904
  for thermocouple, 1084
  half-flash, *904*
  integrating, 902, **912–940**
    as lowpass filter, 940–941
    dual-slope, 902, **914–916**
    multislope, 902, *919*, **918–921**, 921t
    normal-mode rejection, *915*
ADC (cont.)
    powerline rejection, *915*
    single-slope, 902, *914*, **914**
  introduction, 879–880
  micropower, 916t, **941–942**
  noise
    filtering out-of-band, 911
    sampling vs integrating, 940–941
  parallel, **902–907**
  resolution
    effective, 935
    noise-free, 935
  RF
    driving (design example), *906*, 904–907
  selected, 905t
  sequencer, 945
  shootout, 939t
  specialty, 942t
  subsystem, 1084
  successive approximation, 902, 908s, 910t, **908–913**
    charge-redistribution, 950
    design example, *912*, 910–913
  tables, 905t, 910t, 916t, 921t, 935t, 937t, 939t, 942t
  techniques, 902–903
  timing example, 1088–1089
  tracking, 909
  transformer input coupling, 382
  undersampling
    deliberate, 907
  V to *f*, 902, 912–913
adder, *see* logic
admittance, 69, 90
alias, *901*, 931
  digital oscilloscope, 1164
  in DACs, 900
  undersampling conversion, 907
AM radio, 55, 56s
amp, 1
amplifier
  autonulling (design example), 297
    input errors, 306
  balanced audio, 376
  bandwidth vs flatness, 385g
  class-A, 106–107
  class-AB, 108
  class-B, 108
  class-D, 109s
  common-emitter, **87**ff
    biasing, **95**ff
    bootstrapping, 111
    bypassed emitter resistor, 96
    emitter resistor as feedback, 96
    gain of, 94
    input impedance of, 95
    maximum gain of, 98
    nonlinearity, 94, 95, 96s
    revisited (Ebers–Moll), 93
  composite, 332, 543–547, 920
    slew rate of, *920*
  current, *see* transimpedance amplifier
  current-sensing, *278*
  decompensated, 515
  difference, 353t, **347–356**
    as current source, 350–351
    CMRR, 355g
    CMRR trim, 356
    common-mode input range, 354
    filter node, 355
    offset trim, 355
  differential, 375t, **372–380**
    as comparator, 105
    as dc amplifier, 104
    biasing, 104
    BJT, **102**ff
    choosing, **383–387**
    CMRR, 376, 383
    common-mode input range, 376, 383

amplifier, differential (cont.)
compensation, 379
current-mirror load, 105, 153, *496*
distortion, 386g
gain of, 103
gain-setting resistor, 376
high-voltage, *209*, *607*, *694*
input impedance of, 376
JFET, **152**ff
noise, 386, 496, 520
offset voltage, 383
output common-mode voltage, 376
phase splitter, 105
slew rate of, 385
speed, 383
supply voltage, 383
use with ADCs, 380–382
use with single-ended input, 376
equalization, 381, 870–871
error
in linear regulator, 597
in switchmode regulator, *646*, *651*, 651–655
fully differential, *see* amplifier, differential
gain, predictability of, 117
hybrid
low-noise, *534*
input errors, 301
input impedance
effect of feedback on, 118
input protection, 210
instrumentation, *see* instrumentation amplifier
JFET-input, 512
isolation, 583ff, **585**ff
capacitive, 586
JFET, **146**ff
differential, 152
low noise, **509–520**
op-amp hybrid, 152, 153g, 155g, 343–347, 534
series-feedback pair, 150, 151g
with cascode, 148
laboratory, general-purpose, 274
logarithmic, *see* AoE 2nd ed., pp 212–216
noise, *see also* noise
total, vs R_s, 526g
total, vs frequency, 531g
op-amp, *see* op-amp
BJT hybrid, *534*
JFET hybrid, 153g, *545*
low-noise, 522–524t
single-supply, 261
operational transconductance, *see* OTA (operational transconductance amplifier)
output errors, 307
output impedance
effect of feedback on, 119
patch clamp, 552
photodiode, *234*, *548*
amplifier (cont.)
photomultiplier, *843*
piezo, 208, *209*
programmable gain (PGA), 948
programmable-gain (PGA), 370, 371t
sense, 1019
STM, *553*
summing, 234
switchable gain, 182
switching, 109s, 673
transformer in feedback, *285*, *536*
transimpedance, *see* transimpedance amplifier
video and RF, 274, 533
wideband analog link, 353, 380
world's quietest, 505, 507g
amplitude
rms, 1
analog
vs digital, 703
analog switch, 176t
4053-style, 917t, 916–918
charge injection, 176t, 180, 181g
vs R_{ON}, 182g
FET, **171–184**
applications, 182–184
capacitance, 178–181
charge injection, 180
CMOS, 172
feedthrough, 179g
JFET, 172
latchup, 174
limitations, 174
multiplexer, 173
ON resistance, 171, 173g, 175–178
protected, 175
speed, 178
T-switch, *179*
in conversion applications, **916–918**
logic-family, 916–918
transmission gate, *see* logic
analog-to-digital, *see* ADC
aperture time, 940
arbitrary function generator, 18
Arduino, 1086, **1092**, *see also* microcontroller
argument, *see also* abuse, *see* computer
ARM, *see* microcontroller
Art of Electronics
logo wristwatch, 448p
ASCII code, 1039, 1040t
assertion-level logic, 713
asynchronous, *see* logic, *see also* memory
attenuator, 17
matched impedance, 1123, 1124t
audio
analog, 1131
balanced driver, 376
sampling rate, 902
autocorrelation
of PRBS, 978
autozero, *see* op-amp
avalanche, 199, 979, 1069
AVR, *see* microcontroller

B

bandgap, *see* voltage reference
bandwidth
narrowing, **575**ff
lock-in detection, 398, 562, *576*, **575–578**, 851
synchronous detection, 851
of noise, 477
of op-amp, 308
signal averaging, 576
base
spreading resistance, *see* r_{bb}
battery, 2, **686**ff, 688p
backup, 36
characteristics, 687, 689t
choosing, 688
discharge curves, 687g
energy density
vs capacitor, 690g
energy storage
vs capacitor, 690t
Li-ion
charging, *688*
power density
vs capacitor, 690g
primary, 686
rechargeable, *see* battery, secondary
secondary, 686
charging, 687
series resistance of, 9
BCD, *see* number
beast, the number of the, 666
bench instruments favorites, 1152
beta, *see* BJT
bias
tee, 836
biasing
bad, 85
of emitter follower, 83
with compensated V_{BE} drop, 97
with dc feedback, 98
bibliography, **1154**ff
binary
number, *see* number
search, 908
biology
electrical engineer's view of, *805*
biphase code, 1041
bipolar transistor, *see* BJT
bit
effective number of, 928, *929*
grooming, 1138
bitstream, *see* delta-sigma
BJT (bipolar junction transistor), **71**ff
as transconductance amplifier, 91
base resistance, *see* r_{bb}
basic circuit topologies, 90
basic circuits, 91

BJT (cont.)
beta, 72, 74g, 504g, 501–504t
current noise, 483, 493
vs current, 484g
vs frequency, 484g
current ratio, 102g
current sharing in, 112
Darlington, **109**ff, 1081
beta vs current, 110g
Ebers–Moll model, 90
emitter follower, *see* emitter follower
emitter impedance of, 92
gain-bandwidth product, 549g
inductive load, 75
logic with, 124
low-noise, 501–502t
challenge, 507g, **505–508**
replace with JFET, 500
selecting, **500–505**, 507
low-noise design with, **492–509**
noise, 481
from NF, 489–492
model, 488
resistance, 494
test circuit, *557*
vs JFET, 517
paralleling, 112
pinouts, *72*, *502*
power, 106t
pulse generator, 458, 459s
r_{bb}, 481ff, 483g, 488, 501t, 503, 505ff
representative, 74t
rules of thumb, 91
saturation current, I_S, 91
series connection of, 697
simple model, 72
superbeta, 111, 252, 302t, 323, 326
Sziklai connection, **110**ff, 148, 207, 214, 229, 598, 608
table of, 74t, 106t, 501t
terminal notation, 71
V_{BE} tempco, 92
voltage noise, 481, 493
calculation (circuit example), 486
charting, 494
from i_n through R_s, 483
vs r_{bb}, 483g
vs current, 482–483g
vs frequency, 485g
vs power MOSFET and IGBT, 201, 202t, 208t
Black, Harold, 116, 118, *see also* feedback
Blackman–Harris, *see* window, sampling
Bluetooth, 1061
Bode plot, **280–285**, 311, 899, 961, 963
boost, *see* switchmode
bootstrap, **111**ff
cascode, in TIA, 550
leakage cancelation, capacitor, 897
of op-amp follower, 233
bootstrap (cont.)
of power supply, 359
of TIA, 547
twin-T filter, 414
branch
you're sitting on, sawing off, 1091
breakdown, *see also* second breakdown
base-emitter, 82
MOSFET gate, 199
brickwall, *see* filter
buck, *see* switchmode
buffer
logic, 23, 218t, 798g, 817
unity-gain, 311, 843
bus, *see also* computer, 1029t
multidrop, 990
point-to-point, 990
byte, 990

C

cable
cat-5, 864
cat-6, 864
coax, **1116–1123**, **1126**
counterwound helix, 1130p, 1130s
driving with logic, **858–864**, *1120*
reflections, **858–864**, 1116s, 1118–1119s
terminating, **858–864**, **1117**
differential, **864–874**
digital signals through, **856–874**
driving, **856–874**
equalization, 864
pre-emphasis, 864
shielded, 587, *see also* cable, coax
STP, 864
television
analog, 1134
termination
back, 860, **1118–1120**, **1122**
double-ended, 860, 1122
series, 860, **1118–1120**
twisted-pair, **864–874**
UTP, 864
calculus, 1099
CAN bus, **1043**, *see also* computer, data bus, 1061, 1081
capacitance
disappears in terminated cable, 860, 1116
feedback, 113, *see also* Miller effect
gate, 197g
junction, 114
multiplier, 508, *557*, 578, 579g
of analog switch, 178–181
sensor, 1086
capacitive
coupling, 581
isolator, 586
capacitor, **18**ff, 20p
"X" and "Y" types, *see* capacitor, powerline rated
capacitor (cont.)
blocking, 19, 43
bypass, 19, 54
ceramic
ESR and stability, 616
charge pump, 638
current through, 19, 46
dielectric absorption, 28, 211, *298*, 300, 301g, *326*, 422, 915
digital, 1082
electrolytic, 20p, 232, 301g, 392, 580, 597, 633ff, *635*, *661*, 690t, 856
dc coupling eliminates, 496
energy density
vs battery, 690g
energy storage
vs battery, 690t
energy storage in, 19, 686
ESR
filter attenuation, effect on, 393g
increased ripple from, 639
film, *19*, 20p, *279*, 300, 326, 422, 438, 631p
flying
driving MOSFETs, *822*, *847*
in power conversion, *184*, **638**ff
HV discharge circuit, 211
leakage, 300
bootstrap cure, 897
leakage compensation, 298
multiplier, *see* capacitance
mylar, 19, 20p
non-ideal in filter, 392
parallel, 21
piezoelectric effect in, 682
power density
vs battery, 690g
powerline rated, 631, 631p, 664p, 671p
discharge resistor, 631
RC circuit, 21
reactance of, 42, 45, 49g
reactive current, 47g
ripple current, 634, 635s, 659, *see also* current
series, 21
speedup, 808s
storage, 55
ripple current, 634
storage, in power supply, 633
switched, filter, **415–418**
timing, 55
types, 20
variable, 64
CAS, 1019
cascode, 102, 114, 115, *146*, **148**ff, 345, *369*, *377*, *534*, 545, 548, 550
folded, *149*, *377*, *498*, 536
regulated (RGC), *552*
regulator, for HV, *693*, *698*
series, for HV, *697*
catalogs, 1153
cathode ray tube (CRT), 1143

CD audio, 902
ceramic resonator, 450
cesium, *see* oscillator, atomic
chain rule, 1099
charge
  integrator, *see also* AoE 2nd ed., pp 640–643
  integrator (Coulomb counter), *933*
  MOSFET gate, 197g, 198s
  pump, 638, *see also* capacitor, flying, *see also* switchmode
  redistribution
    ADC, 950
    DAC, 909
charge injection
  absence of, 843, 849
  in ADC, 911
  in analog switch, 176t, 180, 181g, 301, 948
    vs R_{ON}, 182g
  in integrator, *918*
  in sample-and-hold, 256
  in switched-capacitor filter, 418
choke, 29, *see also* inductor
  common-mode, 584
cigar, *see* cigar
circuit
  BJT
    basic, 91
  breaker, 630
  diagram
    how to draw, **1101**ff
  integrated, *see* IC
  linear, 14
  loading, *see* loading
  parallel, 2
  RC
    discharge waveform, 21g
    Thévenin simplification, 23
  resistor
    shortcuts, 6
  resonant, **52**ff
  series, 2
    parallel, 5
    resistor, 5
clamp, *see also* SCR
  active, 257
  active reset, in SMPS, 657
  diode, 36
  input transient, in SMPS, 669
  op-amp output, 913
  power supply, 692
clearance, 663
clipping, 81
clock
  generation with PLL, 972
  real-time, 1084
  recovery, 1037–1042
  skew, 757
CML (current-mode logic), 874
CMOS, *see* analog switch, *see* logic
CMRR, 103
  at high-frequency, preserving, 360
  ground-loop isolation, 583
  instrumentation amplifier, 364g
  trim, 360
coax, *see* cable
  digital signals through, 1116–1122
  driving, 1116–1122
coil
  driver
    precision, *898*, **897–899**
Coldfire, *see* microcontroller
common-mode, *see* amplifier, *see* op-amp, *see* comparator
  pickup, 584, *see also* CMRR
  torture test, 869s
common-mode rejection ratio, *see* CMRR
comparator, 24, 236, *see also* Schmitt trigger, 812t, 813t, **809–817**
  window discriminator, *see* AoE 2nd ed., p 669
  cautions, 816
  digital magnitude, 728
  driving logic from, 806–808
  hysteresis
    internal, 815
  input
    common-mode range, 812
  input current, 814, 814g
  output current, 811, 811g
  output swing, 810
  response time, 816g
  saturation voltage, 811g
  schematic, LM393, *270*
  speed, 815
  supply voltage, 815
  voltage
    differential, 815
    offset, 812
compensation
  active
    of op-amp phase error, 314, 315g, 315t
  dominant-pole, 282
  of transimpedance amplifier (TIA), 537, 541g, *545*
  pole-zero, 284
complex
  conjugate, 1098
  numbers, 1097
  plane, 1098
compliance, 8, *see also* current source
components
  best avoided, 63p
  confusing markings, 65p
  itty-bitty, 65p, 269, 627p, 713, 821
  passive, 56ff
  where to buy, 1150
composite amplifier, *see* amplifier, composite
computer, *see also* microcontroller
  architecture, 990–992
    Harvard, 991
    von Neumann, 991
  assembly language, 993, 1086
  BASIC, 1086
  bus-oriented, 991
  C/C++, 1086
  CPU, 990
  data bus, 990, 1029t
    1-wire, 1035–1036
    ADDRESS, 992, 997
    CAN bus, **1043–1045**
    control lines, 992
    DATA, 992, 997
    eSATA, 1037
    Ethernet, 1045
    FireWire, 1042
    GPIB, 1031
    I^2C, *1034*, **1034–1035**
    JTAG, 1036, *1036*
    parallel, 1028
    PATA, 1031
    PC104/ISA, 992
    PCIe, 1037
    PLD address decoder, 1000
    printer port (Centronics), 1031
    SAS, 1037
    SATA, 1037
    SCSI, 1031
    serial, **1032–1037**
    signals, **997–1013**, 1013t
    signals, summary of, 1012
    SPI, *1032*, *1033*, **1032–1034**
    STROBE, 997
    USB, 1042
  direct memory access (DMA), 1010–1012
  flags, 990
  instruction, 990
  instruction decoder, 990
  instruction set
    addressing modes, 994–995
    argument, 994
    subroutine, 996
    x86, **1008**
    x86, simplified, 994t, **993–994**
  interrupt, **1005–1010**
    acknowledgment, 1009
    autovectored, 1009
    handler, 1006–1008
    mask, 1010
    polling, 1008
    shared, 1008
    software, 1010
  Java, 1088
  keyboard bus interface, 1003, 1005
  machine language, 993
  memory, 991, *see also* memory
    cache, 991

computer (cont.)
number storage in memory, 1048
object code, 993
PC104, 997p, 1013, *1014*, 1015p
port, 992
program
counter, 990
example, 996
programmed IO, **998–1005**
command register, 1004
data in, 1001
data out, 998, *999*
read cycle, *1001*
status bit, *1002*
status register, 1002
write cycle, *998*
progress in, 990p
Python, 1088
real-time IO, 992
register, 990, 994–995
SBC, 997p, 1013
software interrupt, 1010
stack, 995
pointer, 990
subroutine, 996
terminology, 989
UART, *see* UART
word, 990
xy vector display, 1000
conductance, **6**
conduction
angle, 635
confusion central, 65p
connector
avoid these, 63p
BNC, 56, 59, 62p
card-edge, 61
circular, 61p
DisplayPort, 1145
DVI, 1145
HDMI, 1144
multipin, 61
photographs of, 60–63p, 1144p
rectangular, 60p
RF and shielded, 62p
shielded-cable, 59
video, **1143**ff, 1144p
converter, *see also* ADC; DAC; power
AD/DA
errors, 881
linearity, **899–900**
performance parameters, 879–880
power
boost, 30
buck, 30
charge-pump, 183
flying-capacitor, 183
correlated double sampling, 553, 570
coulomb, 18, *see also* charge, integrator
counter, *see also* logic
timing with, 465
cPLD, *see* programmable logic device (PLD)
creepage, 665
crest factor, 568, 981
crossover, input, *see* op-amp, rail-to-rail, input crossover
crossover distortion, *see* distortion
crosstalk, 581, 588
crowbar, *see also* voltage regulator
overvoltage, 598, **690**ff, *691*
crystal, *see* oscillator, crystal
CUPL, *see* programmable logic device (PLD)
current, **1**
class-A, 754
comparator input, 814, 814g
comparator output, 811, 811g
critical, in SMPS, *see* switchmode
feedback, *see* feedback, current
input
of digital logic, 795
of instrumentation amplifier, 362
of op-amp, 244
inrush, 661, 668
Kirchhoff's current law, 2
limiting, 597
foldback, 693, 694g, 824, 824g
of outboard pass transistor, 695
load-sensing, 277
loop, 1084
mirror, *see* current mirror
mode
PWM, in SMPS, 652, *see also* switchmode
noise, *see* noise
pulsation in SMPS, 649
reactive, 41
ripple, 634, 635s
in boost converter, 647
in bridge converter, 659
in buck converter, 644
in storage capacitor, 634
in zener bias, 596, 598
sensing, *944*
shoot-through, 186, 660, 760, 760g, 856, 942
shunt, 277, 944
source, *see* current source
spy, 754
transformer, 944
transient, output-stage, 856
current mirror, **101**ff
active load, 105, 153, *496*
compliance, 101, 104
Wilson, *102*, *146*
current source, **85**ff, **620**ff, 623g
3-terminal regulator as, 620
bandwidth of, 254
BJT, 85–88, 146, 534
current source (cont.)
compliance, 8, 27, *28*, 86, *895*
deficiencies of, 87
depletion-mode
FET as, 211, 622, 623g, *696*
difference amplifier as, 350–351
discrete, 622
floating, *895*, 895–897
high voltage, 622, *696*
ICs, 621
instrumentation amplifier as, 367
JFET, 142
JFET and BJT compared, 145
nanoamp
programmable wide-compliance, *896*
pulsed, *555*
nanoamp wide-compliance, 894–897
noise of, **487–489**
op-amp, **228–230**, *242*, 254, *344*, *367*, *623*, *895*
Howland, 229, *230*
oscillation in, *442*
precision, 367, *898*, **897–898**
resistor as, 85
slew rate of, 367
transistor, *see* current source, BJT/JFET/depletion-mode
curve tracer, 1115

D
DAC, *see also* converter, AD/DA, **881–900**, 1061, 1081
1-bit, *see* delta-sigma
application examples, **891–899**
audio, 939t
charge redistribution, 909, *909*
choosing, 891
code-change glitch, 892
current-steering, *884*, **883–886**
generating voltage output from, **885–886**, *886*
delta–sigma, *see* delta-sigma
exemplary, six, *887*, 886–888, 889t
for offset correction, 948
for zero correction, 948
frequency-to-voltage, 890
introduction, 879–880
multiplying, **884–885**, 894t
PWM as, 888
PWM to, converter, 889
$R-2R$, **882–883**, *883*
resistor string, **881–882**, *882*
selected, 893t
tables, 889t, 893, 894t
Darlington, *see* BJT
photo, 842
data bus, *see* bus, *see* computer
data-acquisition system, *see* DAQ
datasheet, too big, 1063

dB, 15ff
dB per octave, 51–52
dc
  blocking capacitor, 43
  power switching, 202
  supply splitter, 262
dc-dc converter, *see* power supply, *see* switchmode
DDS, *see* oscillator, DDS
debugging
  microcontroller, in-circuit, 1092
decibel, *see* dB
decoder, *see* logic
delay equalizer, 415
delay line
  lumped element, 1126, 1128p, *1129*
    impedance of, 1127
  transmission line, 1128p, 1130s
delta-sigma, 902, 915–916
  20-bit ADC, 934
  24-bit ADC, 934–936, *937*
  ADC, 935t, **922–940**
    digital filter in, 924, *926*
  application examples, **932–940**
  audio ADC, 937t
  block diagram, *924*
  charge integrator, 933
  DAC, 930–931, 938–939
  demystifying, **923–931**
  dynamic range, 925, 929
    vs OSR and *m*, 929g
  ENOB, 928, *929*
  fast multichannel IC, 955, *956*
  idle tones, 931–932
    mitigating, 932
    simulation, 933g
  "industrial" ADC, 936, *937*
  magic in the modulator, 927
  microcontroller, implemented with, 932–933
  modulator, *925*, 925–929
    higher order, 928, *929*
  monotonicity, 931
  noise shaping, 927g, **927–931**
  powerline rejection, 954–955, *955*
  pro-audio ADC, 936–938, *940*
  pros and cons, **931–932**
  simplest, 922
  simulation, 929g, 930g, 931g, **928–931**
  spectrum, 931g
  suntan monitor, 922–923
  the paradox, 923–924, 926
  time delay, 931
  vs the competition, **938–940**
derivative, 1099
design review, 1059
Designs by the masters
  instrumentation amplifier, *361*, 359–362
  Keysight's multislope converters, *919*, 918–922
Designs by the masters (cont.)
  nanoamp programmable current source, *896*
  paralleling MOSFETs, *214*
  powerline SMPS, 595p, *666*, 667s, 665–671
  precision DMM, 342–347
  SR560 preamp, *514*, 512–515
desoldering station
  favorite, 1152
detector
  optical, **840–843**
device programmer, 769
dielectric
  absorption, 300–301, 301g
  constant, 18
differential, *see* amplifier
  cable, **864–874**
  pseudo, 584
differentiation, 1099
differentiator
  bridged, 415
  op-amp, 260
  RC, 25, 26s, 51
digital, *see also* logic
  conquest over analog, history of, 703
  filtering, 419
  logic, *see also* gate
  multimeter, 10
  potentiometer, 63
  signal processing (DSP), 418–422
    sampling, 419
  television
    broadcast, 1138
    cable, 1138
    satellite, 1139
    spectrum, 1136s
  vs analog, 703
digital-to-analog, *see* DAC
diode, **31**ff, *32*
  AND gate, 711
  as log converter, 37
  clamp, 36
  current-regulator, 143g
  drop compensation, 35
  forward current, 294–295, 295g
  forward drop, 31
  gate, 36
  I vs *V*, 92g
  inductive clamp, 38
  JFET as, 294
  LED, *see* LED
  limiter, 37
  MOSFET body, 199
  photo, 841p, 841–842
    amplifier, *548*
  PIN, 837p
  protection array, 805
  representative, 32t
  Schottky, 32, 614, 632, 641, 669, 790
  tunnel, 1113
diode (cont.)
  varactor, 64, 440, 960, 969, 971
    capacitance vs voltage, 441g
  zener, 12ff, 82, 674, *see also* voltage reference
    biasing, 675
    drift, 675
    dynamic resistance of, 13
    IC, 676
    low voltage, 13g
    noise, 674, 676
    tempco, 674
    the astonishing LTZ1000, 675
DIP (dual in-line package), *see* IC, packages
direct digital synthesis, *see* oscillator, DDS
discharge
  curves, of battery, 687g
  of high-voltage supply, 211
  RC, 21
discontinued parts, 273 (box)
display, 837–840
  LCD
    interface timing, *1028*
  smart, 839–840
    codes, *840*
  xy vector
    interface, 1000
display devices, 62
DisplayPort, *see* connector
distortion
  crossover, 309
    effect of feedback, 235s
    in push-pull follower, 107
  from slew rate, 386g
  of differential amplifier, 386
  of op-amp, 329–331, 331, 332g
    rail-to-rail, 318
  ultra-low, oscillator, 437
divider
  voltage, 7
DMM, 10
  Agilent's accurate, 342–347
  favorites, 1152
Dobkin, Bob, 604, 679
dogs, 63p
dominant pole
  compensation, 282
droop, *see* integrator
dual slope
  conversion, *see* ADC
dual-tracking, *see* voltage regulator
duty-cycle
  limiting, 465
DVI, *see* connector
dynamic range
  of delta-sigma converter, 925, 929
  of switched-capacitor filters, 418
  vs precision, 292

E
Early effect, **92**
  in current mirror, 101
Ebers-Moll model, **90**ff
edge, *see also* jitter
  detector, 26
  leading, 25
  trailing, 25
EEPOT, 63
egg
  Easter, *see* if you can find all eight
8b/10b code, 1041
8051, *see* microcontroller
electrical engineer
  view of biology, *805*
electromagnetic interference, *see* EMI
electrometer, 297, 570–574
electromotive force, *see* EMF
electrostatic discharge, *see* ESD
EMF, 2
  back, 28
  thermal, 340, 675
EMI
  in instrumentation amplifiers, 365
emitter
  optical, **829–840**
emitter follower, 79–85, *see also* BJT
  as voltage regulator, 82–83
  biasing, 83
  cancelling offset, 85, 87
  example, 84
  JFET current-sink pulldown, 144
  noise of, **487–489**
  output impedance of, 93
  revisited (Ebers-Moll), 93
  with split supplies, 84
e_n-C, *see* transimpedance amplifier
energy
  density
    battery vs capacitor, 690g
  in capacitor, 19
  in inductor, 28
  storage
    battery vs capacitor, 690t
ENOB (effective number of bits), 900, 905t, 928, *929*, 932, 936, 985
equalization, *381*, 864, 870, 871s
equivalent series inductance, *see* capacitor; resistor
equivalent series resistance, *see* capacitor
error
  budget, 293, 295–299
  component, 299
eSATA, *see* computer, data bus
ESD, 200, *see also* human body model; spike input protection, 259, 362
ESL, *see* capacitor; resistor
ESR, *see* capacitor
Ethernet, 867, *see also* computer, data bus, **1045–1046**, 1061, 1081
  8b/10b coding in, 1041
  Manchester coding in, 1040
Euler, Leonhard, 1098
eye diagram, *1028*

F
fan, 627, *see also* AoE 2nd ed., p 858
  proportional control, 608
farad, 18
feature size
  scaling, 794
feedback, **115**ff
  bang-bang, 1070
  BJT examples, 121
  closed-loop gain, 116
  current, **118**, 150, 308, 378, 527
  effect on amplifier circuits, **117**ff
  in PLL, 961
  nonlinear, 276
  nonlinear control, 1075
  output-current sensing, 120
  output-voltage sensing, 117
  PID, 1069, 1074, 1077
    circuit, *1075*
    convergence, *1076*
  split, 264, 285, 913
  stability, 120
    with inductive load, *899*
FET, **131**ff, *see also* JFET, MOSFET
  analog switch, *see* analog switch
  basic circuits, 140
  depletion-mode
    as current source, 622
  dynamic gate current, 164
  enhancement and depletion, 135
  family tree, 136
  G_{max}, 141t
  interdigitation, 140
  JFET and MOSFET, 134
  leakage, circuit cure for, 259
  linear region, 137
  linear switch, *see* analog switch
  manufacturing spread, 138–140
  matching, 139
  saturation current, I_{DSS}, 134, 136, 139g, 361, 847
  saturation region, **137–142**, 147, 158, **165**ff
  similarity to BJT, 132
  square law, 137ff, 165
  subthreshold region, 138, 139g, 166g, *168*
  switch, 132
  tempco of I_D, 138g
  threshold voltage (V_{th}), 137ff, 138g, 147
  transconductance (g_m), 132, 141t, **146**
  transfer characteristics, 135g, 137g, 139g, 142g
  $V-I$ curves, 132, 133g
FET (cont.)
  V_{GS} spread, 140g
FEXT (far-end crosstalk), 581
fiber optics, 853p
  ST/SC, 855
  TOSLINK (EIAJ), 852–853, 854s
  Versatile Link, 853
field-effect transistor, *see* FET
filter
  active, **396–399**, **406–415**
    bandpass, 411–413
    highpass, 408
    lowpass, 241, 401–408
    Sallen and Key, 242g
  allpass, 415
  analog FIR, 983
  anti-alias, 395, 901, 902
  attenuation
    degraded by ESR, 393g
  bandpass
    noise bandwidth, 564t, 565g
  Bessel, 403
    VCVS, 408
  brickwall, 391
  bridged differentiator, 415
  Butterworth, 401, 401g
    LC, **1109**ff, 1110t
    VCVS, 408
  Chebyshev, 401
    phase and amplitude, 400g
    VCVS, 408
    vs *RC*, 396g
  comparison, 402g
  delay equalizer, 415
  digital, 419–422
    in delta-sigma, 924, *926*
    of PRBS, 979
    response, 421g
  eco-friendly, 1073
  effect of component tolerances, 402g
  effect of non-ideal capacitors, 392
  electrically tunable, 412
  elliptic, *403*, 1068
  equiripple, 401
  finite-impulse-response (FIR), 419, 421g
  implementation, **406–418**
  infinite-impulse-response (IIR), 420
  LC, 54g, 54p, 55s, 393
    multisection, 394
  linearity, 422
  lowpass
    dc-accurate, 418
    FIR, 420
    IIR, 420
  maximally flat, 401
  multiple-feedback, 413
    effect of op-amp Z_{out}, 414g
  node of difference amplifier, 355
  noise bandwidth, 564t, 565g

filter (cont.)
notch, 414
passive, **391–396**
performance parameters, 399
phase shift, 404g
phase-sequence, 455, 456g
power-supply, 32
powerline, 631
RC highpass
approximate, 43
exact, 48
RC lowpass
approximate, 42g
exact, 50
phase shift of, 50g
phasor diagram, 52
switchable, 182
RC
loading by, 44
phase shift, 393g
response, 392g
worst-case impedance of, 44
Sallen and Key, 399, 409–410
state-variable, 410–413
bandpass, 411
step response, 405g, 406t
comparison, *406*, 409g
switched-capacitor, **415–418**
sine from square, 435, 436s
Thomson, 403
time delay, 404g
twin-T, 414
types, 400–406
VCVS, 407, 408t
effect of op-amp Z_{out}, 414g
finite state machine (FSM), *see* logic
FIR, *see* filter, digital
FireWire, *see* computer, data bus
first-in, first-out, *see* logic
flash
conversion, *see* ADC
flip-flop, *see* logic
flyback, *see* switchmode
flying
amplifier, 367
capacitor, *see* capacitor, flying
foldback, *see* power supply; voltage regulator
follower, *see* emitter follower, source follower, op-amp
forward
converter, *see* switchmode
FPGA, *see* programmable logic device (PLD)
frequency
analysis of reactive circuits, **41**ff
angular, 14
compensation, **280**ff
in linear regulator, 597
synthesis, 451
frequency (cont.)
synthesizer, 17, 1082
microcontroller, **1065–1069**
to voltage converter, 890
translation, 562
function generator, 17
arbitrary, 18
favorites, 1152
fuse, 630

G
GAL, *see* programmable logic device (PLD)
gate, *see also* logic
delay
vs power, 792g
vs voltage, 794g
logic
as pulse generator, 459
CMOS, 185
speed
vs power, 719g, 792g
vs voltage, 719g, 794g
universal, 715
generalized impedance converter, 398
op-amp bandwidth, effect on, 399g
Gilbert cell, 162
glitch, *see also* logic
code-change, in DAC, 892
glossary, 1166
GPIB, *see* computer, data bus
GPIO, 1084
Gray code, 708
ground, 2
bounce, 758–759, 856
gridded PCB layout, 758
loop, 582
noise, 856
grounding, 579ff
between instruments, 583
blunders, 582
guard, 360, 587
Gummel plot, 92g, 500
Gummel-Poon, 93
gyrator, 397ff

H
H-bridge, *655*, 659, 889
Hamming, *see* window, sampling
Hanning, *see* window, sampling
HDL, *see* programmable logic device (PLD)
HDMI, *see* connector
heat, **623**ff
heatsink, **624**ff, 625p
insulating, 626
LED, 835p
PCB foil as, 626, 627g
sizing, 626g
sizzle test, 625
thermal resistance of, 626g
henry, *see* inductor
heterodyne, 562
Hewlett, William, 437
hexadecimal, *see* number
high voltage, *see also* voltage regulator
clearance distance, 663
creepage distance, 663, 664, 665p
pulse generator, *333*, 917, *918*
hour of power (design example), 466
human
body model (HBM), 200, 804ff
system interface (HI), 1059
humidity
and MOSFET damage, 200
effect on ICs, 684
hysteresis, 237, *see also* Schmitt trigger
in tunnel diode, *1114*
thermal, 683
hysteretic
SMPS, *see* switchmode

I
I^2C, *887*, *see* computer, data bus, 1061, 1081
in DAQ, 954
microcontroller peripherals, **1084–1086**
IC
discontinued, 273 (box)
packages
DIP, 3p, 57p, 65p, 73p, 224p, 225, *269*, 340, 443p, 663, 715p, 720, *758*, 765p, *792*, 861, 1037, 1059
SMT, 73p, 224p, 225, *269*, 340, 443p, 715p, *758*, 765p
parallel interface, 1028, *1030*, *see also* computer, data bus
serial interface, **1032–1037**, *see also* computer, data bus
IEC power entry, 629
IEEE 1394 (FireWire), *see* computer, data bus
IGBT (insulated-gate bipolar transistor), 207
desaturation, *847*
impedance, **40**ff
characteristic, of transmission line, 1116
converter, generalized (GIC), 398
op-amp bandwidth, effect on, 399g
converter, negative (NIC), 397
in parallel, 46
in series, 46
input
of difference amplifier, 352
of differential amplifier, 376
of op-amp, 245
of *LC*, 52
of common-emitter amplifier, 88
of emitter follower, 80
of sources and loads, 79
output
of op-amp, 309
of RRO op-amp, 316

output (cont.)
output, of op-amp, 246
small-signal, 81
transmission-line matching, 1122
index, 1171
inductance
leakage, **632–635**, 656, 668
energy in, 669–670
in powerline transformer, 633
magnetizing, 669–670
of coil, 28
of op-amp output, 250g
sensor, 1084
inductive
kick, 38, *see also* spike
loads, 38
inductor, **28**ff, 29p
flyback, transformer as, 656
reactance of, 44, 45, 49g
stored energy
in switchmode converter, 636
variable, 64p
volt-second balance, 30
input
protection, 804–806, 949
inrush current, *see* current, inrush
instrumentation amplifier, 273, 297, 363t, **356–373**, 1071
as current source, 367
autozero, 370
CMRR, 363t, 364g, 364–365
CMRR trim, 366
CMRR vs R_{source} mismatch, 366g
configurations, 357, 359–360, 368
EMI, 365
in DAQ, 948–950
input current, 362
miscellany, 362
noise, 362
offset trim, 366
PGA, 370, 371t
roll-your-own, 359, *361*
integrated circuit, *see* IC
integrating
conversion, *see* ADC
integrator
droop, 257–260
in active filter, 410
in quadrature oscillator, 453
op-amp, 230, 231s, 257–260
RC, 26, 51
switched-capacitor, 416
interdigitation, 140
interference, 478, 579ff
intersymbol, 870
Internet
video streaming, 1140
interrupt, *see* computer
interrupter
optical, 851
intersymbol interference, 870
inverter
ac power converting, 673s
charge-pump, 183
flying-capacitor, 183
logic
CMOS, 185
op-amp, 225
optional, 232
inverting
switchmode converter, *see* switchmode
IP, *see* programmable logic device (PLD)
IrDA, 1061, 1081
ISA, *see* computer, data bus
isolation
barrier in offline SMPS, 663

J

JFET (junction field-effect transistor), *see also* FET, 217t
$1/f$ noise, 510, 517g
amplifier, **146**ff
common-source, *149*
g_{os} gain error, 167
instrumentation, 512
low noise, **509–520**
maximum gain G_{max}, 167
with cascode, 148
as diode, 294
as variable resistor, **161**ff
linearizing, 161, 162g
capacitance, 170
care and feeding of, 346
characteristic curves, 143g
current noise, **511**
current source, 142
as BJT follower pulldown, 144
self-biased, 143g, 145g
deep subthreshold region, 166g
differential amplifier, 152
dynamic gate current, 164
gate current, 163g
impact ionization, 164g, 546
G_{max}, 141t
hybrid amplifier, 152, 153g, 155g, 343–347
compensation, 154g
vs FET op-amp, 155
in Wien bridge oscillator, 437
leakage, *see* JFET gate current
linear circuits, **142**ff
low-noise, 516t
selecting, **515–517**
mini-table, 141t
noise, 509g, **509–520**
vs BJT, 517
op-amp
fast, 155t
JFET (cont.)
oscillator, 155, 441
in PLL, *395*
spectrum, 442g
output conductance, 166
pinch-off reference, **680**
pinch-off voltage tester, 240
pinch-off voltage, V_P, 136
schematic symbols, 135
series-feedback pair, 150, 151g
source follower, *see* source follower
subthreshold region, 166g, 167g, 168g, **166–169**
threshold voltage (V_{th}), 137, 138g, 147, 150, 166, 167g, 169g
transconductance, 132, 141t, **146**
vs drain current, 168, 168g
vs drain voltage, 170
within a family, 169, 169g
transfer characteristics, **165**ff, 166–167g, 169g
trimming offset with DAC, 345
V_{GS} spread, 140g
voltage noise, 346, **509–511**, 518g
theory vs measurement, 509g
vs transconductance, 509g
voltage reference, **680**
vs MOSFET, 170g
jitter, 427, **457**, 464, 870, 907, 931, 964, 966, 971, 973ff
Johnson noise, 475g, 475t, *482*, 494g, **474–526**, 526g
as noise source, 555ff, 556
calculation example, 486
gain resistor contribution, 152, 352, 358, 376, 906
greater at low gain, 387
in differential amplifier, 520
in JFET channel, **509**
in op-amp, 526
in TIA feedback resistor, **537–550**
joule, 2
joy
and pain, 65
JTAG, 769, 1036, *see also* computer, data bus, 1061
bootloader, 1091

K

Karnaugh map, *see* logic
KCL, KVL, *see* Kirchhoff's laws
Kelvin connection, 277
keyboard
parallel bus interface, 1003, 1005
Kirchhoff's laws, 2, 1107

L

L network, 1123

ladder
R−2*R*, 234
replaced by charge redistribution, 909
lamp, 62
laser diode, 830p, 835s, 834–836, 836s, *973–974*
current, 835s
linearizing, 836s
laser trimming, 244, 305, 352, 357, 361, 416
latch, *see* logic
latchup
in analog switch, 174
LC, *see also* filter, *LC*
Butterworth filter, **1109**ff, 1110t
filter, 54g, 54p, 55s
impedance of, 52
notch, 52
tank, 52
LCD, 1061, 1066, *see* optoelectronics
leakage, *see also* capacitor; JFET, *see also* inductance
spectral, 420
Lebowski, *see* egg
LED, 62, 830p, **829–834**
angular illumination, 832g
backlight, 1084
BJT driver, 76
constant-current drive, 834
desklamp, 835p
display, 837
driver, 1086
forward drop, 76, 827g
lamp (design example), 833
multiplexed display, *751*, 751–752
panel-mount, 832t
pulsed, 466, 834
synchronous detection of, 578
voltage budget, 752
wavelength, 76
Lehrer, Tom, 1099
lightning detector (design example), 497
limiter
diode, 37
line driver
difference amplifier, 351
linear
circuit, 14
linearity
converter, 881, 899
INL vs DNL, 899–900
Little Logic, *see* mini-logic
LMOS, *see* mini-logic
load
capacitive
logic, 857
op-amp, 263s, 264
dump, 618
inductive, 38
load lines, 156g, **1112**ff
loading, 11ff, 11g, 79
lock-in, *see* bandwidth, narrowing
lockup, *see* logic, pathology
logic
0 and 1, 704
active-HIGH, active-LOW, 714
adder, 728
Verilog code for, 729
arithmetic logic unit (ALU), 728, 990
assertion-level, 713
asynchronous, 733
buffer, *see* buffer, logic
capacitive load, 857
clock skew, 757
CMOS, 714
characteristics, **718–720**
circuit, *717*
failure modes, 760
minimizing power, 754
open inputs with, 760
pathology, 760
shoot-through current, 760
threshold skew, 760
weak outputs, 827
combinational, 708
combinatorial, 708
counter, 742t, **741–744**
asynchronous, 733, 734s
modulo-*n*, 748–751
modulo-*n*, timing, 750g
ripple, 733, 734s
synchronous, 737, 737s
data bus, 720, *721*
decoder, 726
Verilog code for, 727
delay
vs power, 792g
vs voltage, 794g
demultiplexer, 726
digital vs analog, 703
divide-by-2, 731
divide-by-3, 734–736, 743s
asynchronous glitch, 743
Verilog code for, 736
driving
external loads, 721, *817*, **817–829**
from comparator, 806–808
from op-amp, 808
driving ac loads from, 821, *822*
driving negative loads from, 821, *822*
ECL, 722
excluded states, 735
expansion, 726
families, 704, 706t
characteristics, **718–720**
circuits, *791*
CMOS, 791–794
ECL, 790
history, 790–794
logic, families (cont.)
input and output, 718
life cycle, 715g
logic levels, 706 (box), 718
RTL, 790
speed and power, 719
speed vs power, 719g
speed vs voltage, 719g
supply voltage, 718
TTL, 790
fan-out, 711, 720, **794–797**
none in transmission gate, 725
FIFO, 747
flip-flop, *196*, **729–733**
clocked, 730
edge-triggered, 731
hold time, 732
master-slave, 732
RS, 729
setup time, 732
toggling, 732
type-*D*, 731
type-*JK*, 731
type-*T*, 731
gate, **708–724**, *see also* buffer, logic; Single Gate
2-input select, 724
AND, **709**
circuits, discrete, 711
circuits, integrated, 717
example, 712, 714
exclusive-OR, 710
interchangeability, 713
inverter, 709
NAND, 710, *see also* memory, nonvolatile
NOR, 710, *see also* memory, nonvolatile
NOT, 709
OR, **709**
standard, 709
transmission, 172, **725**ff
XOR, 710, 737, 753, 771ff, 782, *957*, 975, 983
XOR realization, 723
glitch, 738, 743s
hardware description language, *see also* programmable logic device, *see* logic, HDL
HDL, 711, *see also* programmable logic device
HIGH and LOW, 704
hold time, 732
IC packages, 715p
identities, 722t
input characteristics, **795–796**
input current, 795g, **795**
input overdrive, 806
input protection, 804–806
interfacing, **790–874**
between logic families, *799*, **798–801**

logic (cont.)
Karnaugh map, 723
latch, 740
level, 17
levels, 706 (box)
life cycle, 715g
lockup, 755
magnitude comparator, 728
metastability, 732, 733s, 756
micropower design, 753
mini, 712, 715
minimization, 723
MOSFET, 184
multiplexed display, 751–752
multiplexer, 724
expansion, 726
multiplier-accumulator, 728
negative-true, 714
nMOS
input characteristics, 828
interfacing, 826–829
output characteristics, 827
open-collector, 721
open-drain, 721
output characteristics, **796–797**
voltage vs current, 798g
parity generator, 728
pathology, **755–760**, 760
unspecified rules, 757
positive-true, 714
power
dynamic, 754
minimizing, 754
priority encoder, 727
probing, 808, 809s, 810p
propagation time, 710s
pulse generator
design example, 739
with flip-flops and counters, 739
race, 724, 738, 756, *see also* runt pulse
register, 740, *see also* computer
power, 819t
shift, **744**
removal time, 758
reset/supervisor, 756t
robust link, 1120
RTL, 712
runt pulse, *see* runt pulse
sequential, **729–745**
miscellaneous, 746–748
timing example, 750
setup time, 732
shift register, 1081
feedback PRBS, 976
RAM as, 745
timing, 746g
speed, 710s
vs power, 792g
vs voltage, 794g
logic (cont.)
standard
combinational ICs, 724
gates, 716t
sequential ICs, 740
when to use, 782, 785
startup reset, 755
state machine, 734–737
Mealy, Moore, 736
states, 704, 797g
switch, 172, **725**ff, 916ff, 917t
synchronizer, 737
synchronous, 734
syntax, 711
three-state, 720
through cables, **856–874**, **1116–1122**
comparison, 874
toggling, *196*
transfer function, 796g
transmission gate, 172, **725**ff, 917t, 1016
truth table, 708
TTL, 714
characteristics, **718–720**
circuit, *717*
noise in, 758
output characteristics, 827
unspecified rules, 757
unused input, 718, 738, 754, 759, 806
LVDS trick, 800
wired-AND, 722, 1081
wired-OR, 721, *722*
loss
switching, 662
loudspeaker
as microphone, 486
low-voltage differential signaling, *see* LVDS
LVDS, 705, 864, **868–871**
levels, *873*
power, 874g
signal conditioning, 870, 871s
torture test, 869s
LVPECL (low-voltage PECL), 874

M

magazines, 1153
man
transistor, 75
Manchester coding, **1039–1042**
Mark (RS-232), 871
math, **1097**ff
MDAC, *see* DAC, multiplying
memory, *see also* logic, *see also* computer, *see also* microcontroller, **1014–1027**
cache, 991
computer, 991
configuration, for FPGA, 769
dynamic, 991, **1018–1023**
asynchronous, 1018, *1020*
synchronous, 1021
timing, *1021, 1022*
memory (cont.)
flash, 991
HDD, 992
nonvolatile, **1021–1026**
EEPROM, **1023–1025**, 1053, 1082, 1084
EPROM, 1023
ferroelectric, 1022, **1025**
flash, 1022, **1024–1053**
floating gate, 1022
magnetoresistive, 1022, **1026**
mask ROM, 1023
NAND flash, 1024, *1025*
NOR flash, 1024, *1024*
phase change, 1022, **1026**
PROM, 1023
pseudo-static, 1017
RAM
as shift register, 745
ROM
in state machine, 735
SSD, 992
static, 991, **1015–1018**
asynchronous, 1016, *1016*
synchronous, 1018
timing, *1017*
static vs dynamic, 1015
volatile, 991
volatile/nonvolatile, 1014
wrapup, 1026–1027
metastability, *see* logic
meter, 61
mho, 6
microcomputer
terminology, 989
microcontroller, 747, 749, *887*, **1053–1094**
8051, 1060, *1067*
ac power control (design example), **1062–1065**
circuit, 1063
pseudocode, 1064, *1065*
ARM, 1060
as timer, 469
assembly code
PRBS generator, 779, *783*
AVR, 1060
block diagram, 1054
C-language code
PRBS generator, 782, *784*
constraints, 1086
development environment, **1086–1092**
cost, 1092
development kit, 1090p
downloading code, 1059
frequency synthesizer (design example), 1067p, **1065–1069**
block diagram, 1066
pseudocode, 1067, *1068*
how to select, 1094
implementing delta–sigma ADC, 932–933

microcontroller (cont.)
in-circuit debugging, 1092
in-circuit programming, 1053
loading code on, **1089–1092**
JTAG bootloader, 1091
parallel bootloader, 1091
proprietary bootloader, 1091
SPI bootloader, 1091
UART bootloader, 1090
USB bootloader, 1091
MSP430, 1061
output characteristics, 798g
peripheral ICs, **1078–1086**
direct connection, *1080*, **1079–1082**
I^2C connection, *1085*, **1084–1086**
SPI connection, *1083*, **1082–1084**
peripherals
on-chip, **1061–1062**
PIC, 1060
popular families, **1059–1060**
PRBS generator, **777**ff
Rabbit, 1060
realtime constraints, **1088–1089**
software, **1086–1088**
assembly code, 1086
BASIC, 1086
C/C++, 1086
Java, 1088
Python, 1088
stabilized platform (design example), 1077p, **1077–1078**, 1079p
block diagram, 1078
pseudocode, 1078
suntan monitor (design example), **1054–1059**
C-language code, *1058*
circuit, 1055
pseudocode, 1056, *1057*
terminology, 989
thermal controller (design example), **1069–1077**
block diagram, 1069
circuit, 1071
control loop, 1074, *1075*
pseudocode, *1075*
when to use, 785, 1093
wrapup, **1092–1094**
microconverter, 1070
microphone
loudspeaker as, 486
ribbon preamp, 505
micropower
digital design, 753
microstrip, 858, 1117–1119, *see also* stripline
Miller effect, **113**ff
Millman's theorem, 6, 1108
mini-logic, 715p, 715–719, 725, 772, 800, 916
MiniGate, *see* mini-logic
mixer, 562
modified sinewave, 673s
modulation
amplitude
homodyne, 970
synchronous, 970
amplitude (AM), 55, 56s
BPSK, 970
delta, 910
frequency
demodulation with PLL, 969
Foster-Seeley detector, 970
quadrature demodulation, 969
ratio detector, 970
slope detector, 970
QAM, 970
modulator
delta-sigma, *925*, 925–929
magic in, 927
monostable multivibrator, *see* one-shot
monotonicity
converter, 881
Monticelli output circuit, 318
MOSFET, *see also* FET
avalanche rating, 1073
construction, 134
depletion-mode, 209, 210t
as current source, 622
as input protection, 210
current source, 211
extending regulator V_{IN}, 211, 693
HV discharge, 211
dynamic gate current, 164
floating-gate reference, *see* voltage reference
gate capacitance, 197g, **197**ff
gate driver, 218t, 861
gate threshold, 136
handling precautions, 200
gate damage, 201p
high-voltage
creepage barrier, 665p
logic switch, 184–186
noise, 519
power, **187**ff, 187–191t
applications, **202**ff
body diode, 199
drain capacitance, 198
gate breakdown, 199
gate capacitance, 197g
gate charge, 197g, 198s
gate protection, 199
high-side switching, **202**ff, *204*
lateral, 214g
light-at-night, 206
linear applications, 208
logic-level, 192, *193*
paralleling, 201, 212–215
piezo amplifier, 208
piezo driver, 207
power rating, 199
MOSFET, power (cont.)
protected switch, 205
push-pull follower, 214
source ballasting, 213
switch, 206t
switch examples, 206
switching applications, *195*, *196*, 194–196
switching cautions, 196
thermal runaway in, 213
thermal stability, 187
vs BJT and IGBT, 201, 202t, 208t
R_{ON} vs temperature, 213g
schematic symbols, 134
series connection of, 697
switch, *see also* switchmode
high-side, 825, 826t
protected, 824, 825t
protecting, **823–825**
transfer characteristics, 212g
vs JFET, 170
motorboating, 286
MOV, *see* varistor
MPEG-2, 1137
MPEG-4, 1061, 1137
MSP430, *see* microcontroller
multimeter, 2, 10 (box)
multiplexer, *see also* logic
FET analog switch, 173
in DAQ, 948
multiply
how to, 762
multiplying DAC, *see* DAC, multiplying
multislope
conversion, *see* ADC

N

Nakamura, Shuji, 62
NAND, *see* logic, *see* memory, nonvolatile
negative-impedance converter, 397
negative-true, *see* logic
netlist, *see* programmable logic device (PLD)
NEXT (near-end crosstalk), 581
nice
diagram, *1102*
night light, 206
noise, 15, *see also* op-amp, **473–481**, *see also* amplifier
$1/f$, 476
corner frequency, 566
filtered bandwidth, 564t
forever?, 566, 567s
integrated, 566g
JFET, 510, 517g
op-amp, 324g
analog
filtered PRBS, 975, **977–983**
averaging time, 574
band-limited, 477

noise (cont.)
bandpass filters
response, 565g
bandwidth, 477, **561**ff
multisection *RC*, 561
of *LC*, 562
of an average, 562
bistable, *477*
BJT, *see* BJT, noise
test circuit, *557*
burst, *477*
capacitively coupled, 581
common-mode choke, 584
crest factor, 568, 981
current, **483–484**
measuring, **569**ff
of bias-canceled op-amp, 327, 527
ultra-low measurement, *570*, *572*
current source, of, **487–489**
density, 474, **479**
voltage, **481–483**
differential amplifier, 520
digital
generation with PRBS, **974**ff
emitter follower, **487–489**
e_n-C, 538, 540g, 542s
exceedance probability curve, 574g
excess, 475ff
figure, **479–480**, 480g
contour plot, *490*, *492*
example, 492
filtered
over decades, 567s
filtering out-of-band, 911
flicker, *see* noise, $1/f$
Gaussian, 475g, 979, 980
generator, *559*
pink, *559*
PRBS, 559
immunity, 705
instrumentation amplifier, 362
integrals, 564t
integrated
calculating, 563
of op-amps, 338g, 530, 531g, 531t
interference as, 478
JFET, *see* JFET, noise
Johnson, *see* Johnson noise
low-frequency specification, 564
magnetically coupled, 581
maxima, 574
measurement, **556**ff
hot-cold, 557
transistor test circuit, *557*
Nyquist, *see* Johnson noise
of switching power supplies, 649
of voltage regulator, 618
op-amp, 522t, 523t, 524t, 528g, 529g, 532s, **521–533**
noise, op-amp (cont.)
autozero, **334**ff, 334s
choosing, **525–533**
vs C_{in}, 532g
peak vs rms, 532, 564, 568, 569, 935
popcorn, *477, 478*
potpourri, 574
power supply, **578**ff, 580g, 580t, 913
capacitance multiplier, 508, *557*, 578, 579g
pseudorandom, *see* PRBS
quantization, 574, 927–931
gain, 927
model, 927
resistance, **494**
rms vs magnitude, 574
shaping, *see also* delta-sigma, **927–931**
shot, 327, 476t, **475–476**
source
PRBS circuit, *980*
sources, **558**ff
switchmode power supply, 636g
telegraph, *477*
temperature, **480**
threshold-crossing rate, 574
true random, 982
circuit, *983*
types of, 474, *477*
units of, 474
voltage
from filtered PRBS, 978
voltage reference, **682**, *see also* voltage reference
white, 474–476
zener, 674, 676
nonlinearity
converter error, 881, 899
nonmonotonicity
converter error, 881
NOR, *see* logic, *see* memory, nonvolatile
Norton equivalent circuit, 66, 1108
NRZ, 1041
NRZI, 1041
number
2's complement, 707, 1046
arithmetic in, 708
BCD, 706
codes, **705**ff, 880
floating-point, 708, **1047–1048**
formats, *1047*
Gray-code, 708, *709*
binary conversion, *711*
hex, 705
notation, 705
integer
justification, 1046–1047
signed, 707t
life, the universe, and everything, 42
of the beast, the, 666
number (cont.)
offset binary, 707
sign-magnitude, 707
storage in memory, 1048
nybble, 990
Nyquist
noise, 474

O

offline, *see* ac powered
offset binary, *see* number, codes
Ohm's law, **4**
generalized, 46
not obeyed by diode, 31
OLED, *see also* optoelectronics, 839
one second per hour (design example), 467
1T1C cell, 1018
one-shot, *77*, *242*, *459*, 462t, 463t, **460–465**
555, 460
application examples, 465
cautions, 462–464
problems with, 462–464
protecting pulsed power devices with, 466
retriggerability, 462
timing coefficient, 464g
1-wire bus, *see* computer, data bus
op-amp, **223**ff, 224p
$1/f$ noise, *see* noise
absolute-value circuit, 257
ac amplifier, 226
biasing, 261
active clamp, 257
active peak detector, 254
active rectifier, 238, 257
as voltage regulator, 235
autozero, 272, 295, 335t, **333–342**ff, 370
external, 341
integrated noise, 338g
noise, 336g, 336, 337s, 569t
noise current, 570g
selecting, **338**ff
thermal EMFs and, 340
bandwidth, 247, 249, 308, 328, 329g
basic circuits, 225–231
bias canceled, *125*
better than JFET, 303g
current noise of, 327
capacitive load
stability, 265g
stabilizing, 263s, 264
cautions, 231
chopper-stabilized, *see* op-amp, autozero
clamping, 913
CMRR, 305, 328
CMRR and PSRR, 249
common-mode input range, 245
of autozero, 340
of difference amplifier, 354
compensation, **280**ff

op-amp (cont.)
compensation pin, 320t
current feedback, 270
peaking vs gain-setting *R*, 379g
current source, **228–230**, *242*, 254, *344*, *367*, *623*, *895*
Howland, 229, *230*
current-to-voltage converter, 233
decompensated, 283, *283*, *328*, 542, 550
departure from ideal, 243–249
detailed look, 242–253
die (LT1028), 527p
difference amplifier, 227, **347–356**
CMRR trim, 356
filter node, 355
offset trim, 355
differential input range, 232, 246
differentiator, 260
distortion, 329–331, 331, 332g
slew-rate, 248s
vs frequency, 311g
driving logic from, 808
feedback at dc, 232
follower, 227
with bootstrap, 233
frequency compensation, 247, *see* op-amp, compensation
gain, 249
vs frequency, 281g
gain error, 312g
gain nonlinearity, 312–314
measuring, 313
golden rules, 225
high-power, 272t
high-speed, 310t
high-voltage, 272t
hybrid amplifier with JFET, 152, 153g, 155g, 343–347, 534, *545*
ideal, 243
input
protection, 362
input current, 163, 244, 252, 302, 325
BJT better than JFET, 303g
common-mode voltage, variation with, 304, 304g, 305
effect on integrator, 258
measuring, 325–326
of autozero, 339
temperature, variation with, 303g
input impedance, 245, 250, 301
input offset current, 244, 252
input protection, 259, 362
instrumentation, *see* instrumentation amplifier
integrator, 230, 231s, 257–260
inverting amplifier, 225
JFET,
fast, 155t

op-amp (cont.)
JFET hybrid, *see* op-amp, hybrid amplifier with JFET
limitations, effect on circuits, 249–254
linearity, 251
low input-current, 303t
low noise, 522–524t
choosing, **525–533**
hybrid, *534*
micropower, 273
Monticelli output circuit, 318
noise, 528g, 529g, 532s, **521–533**
0.01–10Hz, $1/f$, $V_{n(pp)}$, 522–523t, 530g, **532**, 564
$1/f$, 528, 529g
autozero, **334**ff, 334s
integrated, 338g, 530, 531g, 531t
low-frequency, 564
vs C_{in}, 532g
noise voltage and current, 249, 323, 324g, *see also* noise, 326
of autozero, 339
of bias-canceled, 327
vs supply current, 525g
noninverting amplifier, 226
nonlinear circuits, 236
offset voltage, 244, 251, 304, 323
effect on integrator, 258
of autozero, 339
test circuit, 323
offset voltage drift, 244
optional inverter, 232
oscillator
precision VCO, 267
output current, 251, 272t
output impedance, 246, 250, 309
effect on MFB filter, 414g
effect on VCVS filter, 414g
vs frequency, 250g, 311, 312g
output range, 231
output swing
vs frequency, 251g, 307g
vs load resistance, 246g
parameters, 245t
unspecified, 296
peak detector
resetting, 255
phase error, 314, 315g, 315t
active compensation, 314, 315g, 315t
phase reversal in, 275
phase shift, 247
photodiode amplifier, *234*
photometer, 265
pinch-off voltage tester, 240
pinout, *225*
power and high-voltage, 272t
power booster, 234
precision, **292**ff, 320–321t
choosing, 319

precision (cont.)
representative, 302t
vs speed, 329
programmable pulse-width generator, 241
PSRR, 306, 328, 533
push-pull booster, 234
rail-to-rail, **315**ff
distortion, 317g, 318
gain, 318g
input crossover, 316, 317g
output impedance, 316
output saturation, 317
output swing, 247g
relaxation oscillator, *see* oscillator, relaxation
representative, 271t
sample-and-hold, 256
schematic
LF411, 243
TLC271, 266
sensitive millivoltmeter (design example), 253, 293, 296t
settling time, 308ff, 320–321t, 328ff
of autozero, 340
sign of feedback, 232
single-supply, 261, 265–270, 322
slew rate, 248g, 251, 328
distortion, 248s
of autozero, 340
vs input differential, 308g
smorgasbord, 232
split feedback, 264
stuck-node tracer, 276g
summing amplifier, 234
supply rail splitter, 262
stability, 262, 263s
supply voltage, 248, 322
of autozero, 338
supply-rail bypass, 232
tables of, 271–272t, 296t, 302–303t, 310t, 320–321t, 335t, 522–524t
track-and-hold, 256
transconductance, *see* OTA (operational transconductance amplifier)
transimpedance amplifier, *see* transimpedance amplifier
triangle oscillator, 239
uncompensated, 283, *see also* op-amp, decompensated
voltage gain, 247
vs frequency, 247g
zero-crossing detector, 269
optocoupler, 805, 837p, 844p, *845*, *846*, *847*, **843–848**, *848–852*
analog, 586, 847
optoelectronics, **829–856**
coupler, *see* optocoupler
detector, 837p, 841p

optoelectronics (cont.)
display
LED, multiplexed, 751–752
displays, 830p, 837p
emitters, 830p
family tree, *829*, 831 (box)
fiber optics, 853p, *see also* fiber optics
image sensor, 841p
interrupter, 578, 844p
LCD, 837p, 839, *840*, 1143
interface timing, *1028*
miscellaneous, 837p
OLED, 1143
optocoupler, 1081
photoconductive, 437
photomultiplier, 837p
PIN diode, 837p
plasma, 1143
television display, 1143
optointerrupter, *see* optoelectronics
optoisolator, *see* optocoupler
oscillation
parasitic, 426, 427s, 442
slow logic transition, from, 801s
oscillator, **425–457**
LC, 439
electrically tunable, 440
555, 430t, **428–432**
50% duty-cycle, 429
full-range duty-cycle, 430
atomic standard, 451
ceramic resonator, 450
CMOS, 426–427
crystal, 443p, **443–450**
a caution, 449
circuits, *447*
micropower, 448, 449g
modules, 449
series and parallel modes, 444, 445g
DDS, 451, *453*, 455, 1066, 1082
emitter-coupled, 440
JFET, 155
low-noise, 441
spectrum, 442g
jitter, 457
local (LO), 562, 971
low-distortion, 437
not needed in μC example, 1056
op-amp
triangle wave, 239
parasitic oscillations in, 426, 427s
phase-shift, 438
PLL, 452
precision VCO, 267
programmable, *267*
quadrature, 454s, 455s, **453–456**
quartz crystal, **443–450**
ratiometric, 431
oscillator (cont.)
relaxation, *267*, **425–435**
CMOS logic, *426*, 427s
op-amp, 425
unijunction, 427
resistor-programmed, 432g
SAW, 450
sawtooth, 430, 917–918, *918*
silicon, 432, 432t
sinewave, **435–443**
Wien bridge, 436–438
TCXO, OCXO, 450
triangle, *239*, *267*, 431
types, 452t
VCXO, **446**, 960, 964, 969, 971
voltage-controlled, *267*, **434**ff, 435, *959*
in PLL, **959–960**
oscilloscope, 2, **1158**ff
analog, *1159*
digital, 1152, **1162**ff, *1163*
aliasing, 1164
dead time, 1164
hints, 1160
probe, 67, 1160, *1161*
skew, 1164
OTA (operational transconductance amplifier), 100
output
comparator swing, 810
current, comparator, 811, 811g
current, op-amp, 251, 272t
output current
critical, in SMPS, *see* switchmode
oversampling, *see* sampling
overtemperature
sensor, 13, 203, 602–604, 604 (box), 671
absent, 695

P

PAL, *see* programmable logic device (PLD)
PALASM, *see* programmable logic device (PLD)
parallel
bootloader, 1091
conversion, *see* ADC
impedances, 46
MOSFETs, 201
resistors, 5
resonant crystal mode, 444, 445g
parameters
unspecified, 296
parity generator, 728
PATA, *see* computer, data bus
PC104, *see* computer, data bus
PCIe, *see* computer, data bus
PCMCIA, 1061
peak detector
active, 254
passive, 254
PECL (positive ECL logic), 874
permeability, 29
PGA, *see* amplifier, programmable gain
phase
detector, 575, *see also* PLL
equalizer, 415
error
of op-amp, 314, 315g, 315t
sequence filter, 455, 456g
splitter, 105
with discrete BJTs, 88
phase shift
and feedback stability, **280–285**, *see also* Bode plot
filter comparison, 404g
oscillator, 438
RC lowpass, 50g
phase shifter
constant-amplitude, 88, *89*
phase-locked loop, *see* PLL
phasor, 51
photodiode, 841–842, *see also* diode, photo
photometer
single-supply, 265
photomultiplier, *842*, 842–843
amplifier, *843*
photoresistor, *206*, 844, 845p
phototransistor, 841–842
pi network, 1109, 1124
PIC, *see* microcontroller
pickup, 442
common-mode, 584
PicoGate, *see* mini-logic
PID loop, *see* feedback
piezo, 24
piezo driver, 207, *332*
piezoelectric effect
capacitor, in, 682
PLA, *see* programmable logic device (PLD)
platinum RTD, *see* sensor, temperature
PLD, *see* programmable logic device (PLD)
noise generator, 981
output characteristics, 798g
PLL, 452, 972t, **955–974**, 1071
AM demodulation, *970*
applications, **966–974**
block diagram, *956*
BPSK demodulation, *971*
capture and lock, **964–965**
clock generation, 972
“cut and try”, 963
design, **960–964**
FM demodulation, *969*
frequency multiplier (design example), **961–964**
circuit, *962*
frequency synthesis, **966–969**
fractional-*n*, *966*, **966–967**
rational-approximation, *969*, 968–969

PLL (cont.)
laser offset locking, *973*, 973–974
loop
block diagram, *961*
loop filter, 961
loop gain, 962, 963 (box)
loop stability, 961
phase detector, *957*, 958s, **956–959**, 959s
backlash, 958–959
dead zone, 958–959
pulse synchronization, 971
rejection of noise and jitter, 974
sampling clock, 907
signal regeneration, 971
spur removal, 394
VCO, **959–960**
4046-type, *959*, 959s
with JFET oscillator, 394
PMT, *see* photomultiplier
pod, 1091, 1093
polar coordinates, 1097
pole, 52
popcorn
noise, 477, *478*
positive-true, *see* logic
pot, *see* potentiometer
pot core, 29
potential difference, *see* voltage
potentiometer, 7, 8p
digital, 63, 184, 412, 1082
power
ac detect, 1081
ac monitor, 1082
an hour of (design example), 466
density
battery vs capacitor, 690g
design, **623**ff
dynamic, 754
factor, 47
correction in offline SMPS, 668
in offline SMPS, 661
in unregulated supply, 635
in reactive circuits, 47
in resistor, 6
micro, *see* micropower
minimizing, in CMOS design, 754
one minute of, 23
precision source, 285
semiconductor packages, 628–629p
spectrum
from PRBS, 979
of SMPS, 636
supervisor, 1079
switching
with MOSFETs, 192–200
with outboard driver, 237
tradeoffs
noise, 483g, 509g, 525g
speed, 792g
power (cont.)
train, *see* switchmode (SMPS)
transfer, 11
transistor, 624
series connection, 698
power supply, **594**ff, 595p, *see also* voltage regulator
clamp, 692
commercial modules, 684, 685p
current limit, 597
foldback, 693, 694g
of outboard pass transistor, 695
favorites, 1152
filtering, 32
high voltage, **695**ff, *696*
linear
compared with switchmode, *see also* switchmode, 635–638
topology, compared with SMPS, *630*
noise, *see* noise, power supply
offline, *see* switchmode, offline
overvoltage protection, 598, **691**ff
rectifier, 634
damping spikes, 634
regulated, 123
regulator, 34
resonant charging, 39
ripple voltage, 33
split, 34
storage capacitor, 633
ripple current in, 634
switchmode, **636**ff, *see also* switchmode
ac-powered, *see* switchmode, offline
compared with linear, *630*, 635–638
controller ICs, 658t
current-mode ICs, 654t
programmable, 1086
voltage-mode ICs, 653t
unregulated
bench vs SPICE, 634–635
from ac powerline, **628**ff
power factor in, 635
voltage multiplier, 34
zener as, 595
powerline
capacitor, 631
discharge resistor, 631
entry, 629
filter, 631
monitor IC, *943*, 943–944
rejection in integrating ADC, 915
switch, 631
transformer, 632
leakage inductance in, 633
rating, 633
PRBS, **974–982**
analog noise from, 975, **977–983**
analog noise voltage, 978–981
chipping sequence, 975
PRBS (cont.)
digital
filter, 979
eye diagram, 871s
feedback shift register, *976*
properties, 976
taps, 976t
noise generator, *559*, *771*
spectrum, 560s
noise source
circuit, *980*
power spectrum, 979
with cPLD, 772
with discrete logic, *559*, *771–772*
with ECL logic, 981, 983p
with microcontroller, **777**ff
assembly code, 779, *783*
C-language code, 782, *784*
with PLDs, 981
wrapup, 981
pre-emphasis, 864, 870, 871s
precision
vs dynamic range, 292
prefixes, 4 (box)
printed circuit
SMT vs through-hole, 268, *269*
printer port (Centronics), *see* computer, data bus
priority encoder, *see* logic
probe
for logic signals, 808, 810p
probing
digital signals, 808, 809s, 810p
programmable (with voltage)
current source, *228–230*, *367*, *896*
oscillator, *267*
power supply, *608*, *610*
pulse generator, *918*
timer (one-shot), *242*
voltage source, *195*, *892*
programmable logic device (PLD), 728, 737, 745–746, **764**ff, *see also* logic, 765p
ABEL, 764, 775, *776*
address decoder, 1000
advice, **782**ff
behavioral coding, 777
cPLD, 764, 766
PRBS bit generator, **772–777**
CUPL, 764
FPGA, 764, **768–770**, 1031
configuration memory, 769, 770
logic element, *770*
GAL, 764
HDL, 766, *see also* logic
HDL entry, **775**ff
mixed-signal, 769
netlist, 769
packages, 765p

programmable logic device (cont.)
PAL, 764, **765–768**
circuit, *766–767*
macrocell, 764, *768*
PALASM, 764
PLA, *768*
programmer, 769
schematic entry, **773**ff, *774*
software, 769
structural coding, 777
Verilog, 766, 775, *778–781*
VHDL, 766, 777
when to use, 782ff, 785
programmable oscillator, *267*
pseudo-differential, 275, 584
pseudo-random bit sequence, *see* PRBS
PSoC, *see* microcontroller
PTAT, *see also* bandgap, 92, 104, 431, *606*, *680*, **679–680**
pulse, 16
duty-cycle
limiting, 465
forming network, 1126
from step, *77*, 459s, 458–468
generator, 17
high-voltage, *333*, 917
n-pulse, 752–753, *753*
programmable pulse-width, 241
single-pulse, 739
triggerable (design example), 739, *740*
with discrete BJTs, 77–79
with flip-flops and counters, 739
limiting width of, 465
runt, *see* runt pulse, *see also* logic
synchronization
with PLL, 971
width modulation, *see* PWM
push-pull
BJT output stage, 106–108
biasing, 107
crossover distortion, 107
thermal stability of, 108
op-amp power booster, 234
PWM, *see also* switchmode, 1061, 1071, 1077
as digital-to-analog converter, 888
in switchmode converters, 645ff
light dimming, *834*
rate multiplier as, 890
thermal controller, **1069–1077**
to DAC converter, 889
torque-motor control, *890*
pyroelectric
detector, 841p

Q

Q (quality factor), 52
QAM, 456, 1138
quadrature
encoder, 852, 877
quadrature 9cont.)
FM demodulation, 969
modulation, 577
oscillator, 454s, 455s, **453–456**
quadrature-amplitude modulation, *see* QAM
quantization
noise, 927–931
gain, 927
model, 927

R

R−2*R*, *see* ladder
Rabbit, *see* microcontroller
radio
AM, 55, 56s
radiofrequency, *see* RF
rail splitter, 262, 277
rail-to-rail, *see* op-amp
ramp, 15
generator, 27
random walk, 981
RAS, 1019
raster, 1132
rate multiplier
as PWM, 890
r_{bb}, *see* BJT (bipolar junction transistor)
reactance, **40**ff
graph of, 49
reactive circuits, **40**ff
power in, 47
reading and references, **1154**ff
rectification, **31**ff
rectifier, **31**ff, 634
active, 238
bridge, 32
fast-recovery, 670
forward drop, 31
full-wave, 32
active, 257
full-wave center-tapped, 33
half-wave, 32
power supply
spikes, 634
signal, 35
reference design, 671
refresh, *see* memory, dynamic
register, *see* logic
regulation
secondary, in SMPS, 671
regulator, *see* voltage regulator
relay
driving from logic, 817–820
electromechanical, 59
optocoupled, 843–851
solid-state, *see* SSR
reset
logic startup, 755
of core, in SMPS, 657
resistance
dynamic, 12
internal, 11
leakage, 10
negative, 1113
of analog switch, 175
small-signal, 12
source, 11
thermal, *see* thermal resistance
resistor, **3**ff, 3p, 5 (box)
accuracy, 1106
chip
sizes, 4
color code, 1105t
digital, 1082
discharge, for powerline capacitor, 631
E96 series, 1104t
emitter-ballasting, 113
in parallel, 5
in series, 5
nonlinearity, 698, 1106
power in, 6
properties of, 300
standard values, 1104
tempco, 1106
types, **1104**ff, 1106t
variable, 63
JFET, 161
voltage coefficient, 698, 1106
wirewound, 3p, 8p, 63p, 591, 1070, 1105
resonant
charging, 39
circuit, **52**ff
RF
driving cable, 862
RFI (radiofrequency interference), 582
ribbon microphone, 505
ringback, 759
ringing
in SMPS, 668
ripple, 635, 636s, 639g
current, 644
effect of capacitor ESR, 639
effect of switchmode frequency, 641
filtering, 619
in unregulated supply, 599, **632**ff
rejection, 605
rise time, 16
RLL, 1041
Rogowski coil, 944
RS-232, **871–874**, 1039p, **1038–1039**, 1081
and cable length, 872s
driver ICs, 872
levels, *873*
power, 874g
pseudo-random byte generator, 771
rates, 873g
signals, 1039t
timing ambiguity in, 1038

RS-422, 865s, 866s, **865–867**
levels, *873*
power, 874g
RS-485, 865s, 866s, **865–867**, **1038–1039**
isolated, *868*
levels, *873*
power, 874g
RTC, 1084
RTD, *see* sensor, temperature
rubidium, *see* oscillator, atomic
runt pulse, 737s, *738*, 743s, 748, 753, *see also* logic race, 757

S
S/PDIF, 1061
safe operating area (SOA), 192g, 216, 627
Sallen and Key filter, 399, 409–410
sample
PLL clock, 907
sample-and-hold, 183, *see also* op-amp
sampling, 419, *901*, **901**ff
bandpass, 907
depth
and dynamic range, 900
direct IF to digital, 907
Nyquist criterion, 900, *901*
over, *see also* delta-sigma, 901, *901*, 923
ratio (OSR), 924
rate
and aliasing, 900, *901*
and bandwidth, 900
rate and aliasing, 931
under, *901*, **901**ff
band conversion, 907
SAS, *see* computer, data bus
SATA, *see* computer, data bus
saturation
in BJT, 73
of magnetic core, 657, 659
sawtooth, *see* oscillator
schematic
diagram
how to draw, **1101**ff
entry
in SPICE, 1146
Schmitt trigger, 237, 269, 459, 717, 754, 801s, 803s, 807–809s
speedup capacitor, 808s
with discrete BJTs, 79
SCR
latchup, 759
overvoltage crowbar, 598, **691**ff
SCSI, *see* computer, data bus
SD card, 1061, 1082
second breakdown, 192, 216, 627
Seebeck effect, *see* thermal EMF
sensor, *1079–1086*
ac power, *943–944*
atom, 553ff
sensor (cont.)
capacitive, *505*
charge, *933*
current, *278*, *943–944*
grasp, *372*
light, *497*, 539ff, *547*, *549*, 578, 837p, 841p, 840–843, 844p
lightning, 497
sound, 486, 505ff
strain, 297ff, 588, *937*
suntan, **278–280**, *922*, **1054–1059**
temperature, **1070**
RTC, 1069ff, *1071*
RTD, 347, 1070
semiconductor, 621, 680
thermistor, *123*, *372*, 608, 1070
thermocouple, *see* AoE 2nd ed., pp 988–992, 339, 341t, 367, 675, *937*, 1070, 1084
SERDES, 855, 869
serial, *see* computer, data bus, *see* logic
series
impedances, 46
resistors, 5
resonant crystal mode, 444, 445g
termination, **858**ff
series-feedback pair, 122
with JFET, 150, 151g
servo, 1069
settling time, 309g
of op-amp, 308ff, 320t, 328ff
setup time, *see* logic
SH-4, *see* microcontroller
shielding, 579ff
shift register, *see* logic
shoot-through, *see* current
shootout
ADCs, SAR vs ΔΣ, 939
at the ADC corral, 939t
JFET vs BJT, 517
low-power ADC, 941–942
shunt
current, 277
siemens, 6, 90, 147
signal
averaging, 576
generator, 17
guard, 360, 587
mixed
PLD, 769
regeneration
with PLL, 971
sampling, 419
slew rate of, *362*
to-noise ratio, *see* SNR
signals, **14**ff
silicon-controlled rectifier (SCR), *see* SCR
SIM Card, 1061
simulation, *see* SPICE
sinewave, 14
from square wave, 394s, 435
inverter, 673s
modified, 673s
oscillator, *see* oscillator, sinewave
Single Gate, *see* mini-logic
single pulse, *739*
single slope
conversion, *see* ADC
6T cell, 1016
sizzle test, 625
skin depth, 1122g
slew rate, *see also* op-amp
and jitter, 457
causes interference, 581, 588, 705
filtered, *1073*
in oscillators, 426
of composite amplifier, *920*
of current source, 367
of differential amplifier, 385
of op-amp, 248, 308, 328, 330, 340, 515, 862
of signal, *362*
reduced, 607, 650, 759, *820*, 865, 871
too slow, 760, 808
small-signal, 12, 45, **80**, **90**, 147, 583
impedance, 81, 245
symbol for, 45, 80, 90, 1097
transistor, 74t
SMU, 554, 1115
favorites, 1152
SNR, **478–481**
snubber, *39*, *630*, 634, 656, 660, 669
soft start, 606, *607*, 650
solar cell, 2
soldering iron
favorite, 1152
solenoid, 29
source
floating, for bias, *215*
sinewave, 41
source follower
JFET, **156**ff
gain, 157
input impedance, 157
low distortion, 160, 161g
output impedance, 157
tutorial, *159*
with active load, 158
push-pull power MOSFET, 214
source-measure unit, *see* SMU
source-measure unit (SMU), 897
Space (RS-232), 871
SparkFun, 1082
spectrum
analyzer, 567
SPI, *887*, *see also* computer, data bus, 1061, 1081, 1084, 1086
ADC SAR example, 909

SPI (cont.)
bootloader, 1091
in ADC, *936, 937*, *943*, *945*
in DAQ, *946, 947*, 951, 954
isolation in DAQ, 950–951
microcontroller peripherals, **1082–1084**
SPICE
amplifier analysis, 314, *315*
comparison with bench, 150, 459s, 635s
crystal analysis, 443–446
filter analysis, *393*, *399*, *414*, *456*, *560*
primer, **1146**ff
simulation
comparison with bench, 1148s
unregulated power supply
bench comparison, 634, 635g
spike, *see also* charge injection, 17
differentiated edge, 25, 26s
from discharge, 200
from flyback, 39, 193
from leakage inductance, 669
from snapoff, 199, 634
ground-current, 856, 857
in frequency domain, 977
in power supply rectifier, 634
in SMPS driver, 669
logic driving capacitive load, 857
spur
removal, 394
square wave, 16
converting to sinewave, 394s
quadrature, 455
SSR, 24, 59, 848, **848–850**, *850–851*, 1062, 1081
stabilized platform
microcontroller, 1077p, **1077–1078**
standing-wave ratio, *see* VSWR
state machine, *see* logic
state-variable filter, 410–413
step, 17
response
filter comparison, *406*, 409g
stepping motor, 1084
STM (scanning tunneling microscope), **553**ff
image of Si crystal, 553p
STP, 583
stripline, 456, 858, 864, 871s, 1117, *see also* microstrip
stuck node
tracer, 276g
successive approximation, *see* ADC
suntan monitor
delta-sigma, 922–923
microcontroller, **1054–1059**
op-amp, current divider, 279
op-amp, integrating, 278
op-amp, two-step, 279
with ADC, *922*
with microcontroller, **1054–1059**
superbeta, *see* BJT
surface acoustic wave, *see* oscillator, SAW
surface-mount technology (SMT), 65, 66p, 627p
susceptance, 90
switch
3-way, *59*
analog, *see* analog switch, 948
multiplexer, 948
as input device, 802–804
bounce, 802, 803s
break-before-make, 179
debouncer
latch as, 730
SR flip-flop as, 730
debouncing, 729, 802
DIP, 57p
dry-switching, 58
floating, 203
high-side, 821, 825, 826t
matrix, 1069
mechanical, 56, 57p
MOSFET, *see* switchmode; MOSFET
opto-interrupter, 851
PCB-mounting, 57p
powerline, 631
protected, 175, 824, 825t
protecting, **823–825**
pushbutton, 58
reed, 573
rotary, 58
simple logic with, 58
T-switch, *179*
toggle, 57p
toggling, 196
switch, analog, *see* analog switch
switched-capacitor
filter, **415**
integrator, 416
resonator, 454
switched-capacitor filter, **418**
switching
amplifier, 109s, 673
loss, 662
switchmode
zero-current switching (ZCS), 823
zero-voltage switching (ZVS), 823
switchmode (SMPS), *see also* voltage regulator, **636**ff
boost, **647**
bridge converter, **659**
buck, **642**ff
buck, examples, 644ff
buck-boost, **651**
compared with linear, *630*, 635–638
continuous-conduction mode, 644, 670
controller ICs, 658t
core, gapped, 656, 670
critical conduction mode, 643
switchmode (cont.)
critical output current, 643
current pulsation, 649
current-mode ICs, 654t
design equations, 644, 647–648, 656, 657, 659
efficiency, 636
flyback, **655**
power train, 671
transformer in, 670
forward converter, **656**
H-bridge, *655*, 659
hysteretic, 644
inductorless, **638**ff, 639g, 639t
regulated, 641
ripple, 639, 641
internally compensated, 645–646
inverting, **648**
isolated, **653**ff
flyback, 655
topologies, *655*
loss, 662
low-noise, 649
noise, 649
noise spectrum, 636g
non-isolated, **641**ff
boost, **647**
buck, **642**ff
buck, examples, 644ff
buck-boost, 651
design equations, 644, 647–648
hysteretic, 644
internally compensated, 645–646
inverting, **648**
waveforms, *643*, *647, 648*, *652*
wrapup, **649**ff
offline, **660**ff
dual-voltage, 660
inrush current, 661, 668
power-factor correction (PFC), 661, 668
real-world example, **665**ff
when to use, 672
wrapup, 672
point-of-load, 637, 686
PWM modes, 651
soft start, 650
topologies, 638
compared with linear, *630*
voltage mode, 652
voltage-mode ICs, 653t
waveforms, *643*, *647, 648*, *652*, *666*, 667s
wrapup, non-isolated, **649**ff
zero-current switching (ZCS), 650, 662, 850, 1062
zero-voltage switching (ZVS), 650, 659, 671, 850, 1062
synchronous, *see also* logic, *see also* memory
synchronous detection, *see* bandwidth, narrowing

synthesizer
clock, 1084
Sziklai, *see* BJT

T
T network, 1109, 1124
tables, *see* Table of Contents
take-back-half, 1075
convergence, 1076g
television
analog
broadcast, 1134
cable, 1134
recording, 1135
spectrum, 1136s
cable
switched broadcast, 1142
digital, **1136**ff
broadcast, 1138
cable, 1138
recording, 1142
satellite, 1139
spectrum, 1136s
satellite
uplink, 1140p
tutorial, **1131**ff
temperature
coefficient, *see also* referent, e.g., voltage reference, etc.
stabilized
oscillator, 450
voltage reference, 681
temperature coefficient
ways to specify, 683g
temperature controller, *see* thermal controller
temperature sensor, *see* sensor
terminal nomenclature, 131
termination
of transmission line, **858**ff, 1117
Thévenin equivalent circuit, **9**ff
examples, *1107*
Thévenin's theorem, **1107**ff
generalized, 55
thermal controller, 123
microcontroller, **1069–1077**
thermal EMF, 339–340, *see also* op-amp, autozero, *see* EMF, 675
thermal hysteresis, 683
thermal resistance, 216, **624**ff
of heatsinks, 626g
of PCB foil, 627g
transient, 628, 1074
thermal runaway, *see also* second breakdown, 108ff, 112, 187, 212ff, *214*
absent in MOSFET, 138, 215ff
thermistor, *see* sensor
thermocouple, *see* sensors, temperature
thyristor, 208
TIA, *see* transimpedance amplifier
time
delay
filter comparison, 404g
delay circuit, 23
hold, *see* logic
propagation, *see* logic
removal, *see* logic
setup, *see* logic
time constant, **22**
timer
an hour of power, 466
camera control (design example), 467
ICs for, 467
microcontroller as, 469
one second per hour, 467
programmable, 242
timers, **457–469**
555, 460
TinyLogic, *see* mini-logic
tolerance, 6, 300
effect on filter response, 402
tools
favorites, 1152
toroid
common-mode choke, 585
totem-pole, 695
touchscreen, 944–945, 1082, 1084
track-and-hold, *see also* op-amp, 911
tracking ADC, 909
transconductance, 89, *see also* BJT, FET, JFET
transconductor, *see* OTA (operational transconductance amplifier)
transducer, *see* sensor
transfer function, 7
transformer, **30**
for wideband ADC input, 382
in feedback loop, 285, 286g, *536*
isolated SMPS, **653**ff, *see also* switchmode
powerline, 632
leakage inductance in, 633
rating, 633
signal, **535–537**
signal isolator, 586
variable, 64p
transient
capture, in PLL, 964
thermal resistance, 628
transient response
of voltage regulator, 616
transimpedance amplifier, 233, **537**ff, *548*
bandwidth, 537–539
bootstrapped cascode, 550
bootstrapping, 547
capacitive feedback, 552, 570
cascode input, **548**ff
commercial, 570
composite, **543**ff
current calibrator, *555*
transimpedance amplifier (cont.)
noise, 497
e_n-C, 538, 540, 541g, 542s, 544s
input, 538
patch clamp, 552
regulated cascode (RGC), 551
stability, 537
STM, *553*
world's best, *545*
transistor
bias-resistor, 712
bipolar, *see* BJT
BJT switch, 73–79
diamond, 100
digital, 712
field-effect, *see* FET
man, 75
packages, 73p
pass, 594, 695
protection, 695
perfect, 99
photo, 841p, 841–842
pre-biased, 712
series connection of, 697
thermal resistance of, 624, *see also* thermal resistance
unijunction, 427
transmission gate, *see* logic
transmission line, **1116**ff
and sinewaves, **1120**ff
characteristic impedance, 1116
delay line, 1128p
differential, 864–873
LVDS, 868–873
digital signals on, **858–873**
power dissipation, 874g
impedance matching, 1122
broadband lossless, 1124
minimum loss, 1123, 1124t
narrowband lossless, 1125
loss, 1121, 1122g
matched
capacitance disappears, 1116
reflection, 1117, 1118s
resistive attenuator, 1123, 1124t
ringing, 1119s
termination, 860–864, **1117**ff
back, 1118
series, 1118
twisted pair, 1117
unterminated, 859s, 860s, 858–860, 1119
transresistance amplifier, *see* transimpedance amplifier
triangle wave, 15
triboelectric effect, 573
trigonometry, 1097
truth table, *see* logic
TTL, *see* logic

TVS, 39
twin-T filter, 414
twisted pair
  wideband analog over, 352
2's complement, *see* number, codes

U
UART, *see also* computer, **1038**ff, 1061ff, *1071*, 1089
  bootloader, 1090
ugly
  diagram, *1102*
undersampling, *see also* ADC, *see* sampling
uninterruptible power supply (UPS), 673
unused input, *see also* logic
  in ADC, 935
UPS, *see* uninterruptible power supply
USB, **1042**, *see also* computer, data bus, 1061, 1081
  bootloader, 1091
  to RS-232 adapter, 871

V
V to *f*
  conversion, *see* ADC
varactor, 64, 960, 969, 971
  tuned oscillator, 440
Variac, 64p
varistor, 39
VCO, *see* oscillator, voltage-controlled
Verilog, *see also* logic, *see* programmable logic device (PLD)
VFD, 839
VHDL, *see* programmable logic device (PLD)
Victoreen, 3p, 698
video
  analog, 1132
  composite, 1133s
  connectors, **1143**ff, 1144p
  digitizer, 1084
  modulation, 1133
  over Internet, 1140
volt, 1
voltage, **1**
  as complex quantity, **44**ff
  burden, 10
  common-mode, *see* amplifier
  comparator output swing, 810
  compliance, 8
  converter
    flying capacitor, 183
  divider, 7
    adjustable, 7
    equation, 7
    generalized, 48
  doubler, *see also* charge pump, 34, *639*, 660, 872, 1063
  drop
    of diode, 31
voltage (cont.)
  dropout, *see* voltage regulator
  high, *see* high voltage
  Kirchhoff's voltage law, 2
  mode
    PWM, in SMPS, 652, *see also* switchmode
  multiplier, 34
  noise, *see* noise
  offset, *see also* amplifier; op-amp
    comparator, 812
    correction with DAC, 948
    op-amp, 244
  offset in op-amp, *see* op-amp
  power-supply ripple, 33
  reference, *see* voltage reference
  regulator, *see* voltage regulator
  signal, 14, 17, 21–27, 35–37, 43–45, 49g
  source, 8
    8-channel, 893, *895*
    low-noise, 891, *892*
  source, general laboratory, 891, *892*
  spike, *see* spike
  thermoelectric, 341t
  threshold
    of digital logic, *see also* logic, 796–797
  triboelectric, 573
voltage reference, **674**ff
  2-terminal, 677t
  3-terminal, 678t
  accuracy, 683
  bandgap, 104, 599, 602, 604 (box), *606*, 611, 677, 678t, **679**
    as temperature sensor, 680
    noise in, 679, 681, 682
  current source with, 895
  drift, 683
  humidity, 684
  in DAQ, 949–950
  JFET pinchoff, **680**
  MOSFET floating gate, **681**
  noise, 682g, **682**
  precision, 681
  PTAT, *see* PTAT
  self-heating, 684
  series, 678t
  shunt, 677t
  tempco, 683
  V_{BE}, *see* voltage reference, bandgap
  x-ray exposure, 684
  zener, 674
    biasing, 675
    bootstrap biasing, *676*
    compared with bandgap, 679
    drift, 675
    dynamic resistance, 674
    IC, 676, 677t
    LTZ1000, the spectacular, 675
    noise, 674, 676
voltage reference, zener (cont.)
    tempco, 674
voltage regulator
  3-terminal
    as current source, 620
    floating HV, 698
  317-style, 604 (box), 605t
    ADJ-pin current, 605
    as current source, 620
    circuit, *606*
    dual voltage, *612*
    dual-tracking lab supply, 608
    high-voltage supply, 609
    hints, 604, *607*
    proportional fan control, 608
    schematic, *603*, *606*
  7800-style, 602t
  723, **598**ff
    in defense of, 600
  current limit, 693
    foldback, 694
  current-reference, 611
  emitter follower as, 82–83
  extending V_{IN}, 211, 693
  fault protection, 619
  filtering, 619
  fully integrated, **600**ff
  high voltage, *609, 610*, **695**ff, *696*
  linear
    3-terminal adjustable, **602**ff, *see also* 317-style
    3-terminal fixed, **601**, 602t
    bypassing, 616
    choosing, 613
    crowbar, 598
    current limit, 597
    dropout voltage, 606g
    error amplifier, 597
    feedback stability, 597
    ground-pin current, 616
    high-voltage, 693, 698
    idiosyncrasies, 613
    LDO stability, 616
    noise, 618
    pinouts, 613
    reverse polarity, 616
    taxonomy, 601
    transient response, 616, 618s
    tutorial, **595**ff, *596*
  load dump, 618
  low dropout, 599
  low-dropout (LDO), **610**ff, 612g, 614–615t
  LT3080-style, 611
    as current source, 620
  op-amp, 235
  ramp-up control, 606
  schematic, LM317, *603*, *606*
  switching, 609

voltmeter, 2
  sensitive (design example), 253, 293
VOM, 10
VSWR, 1121

W
wall plug
  blades and wires, 628ff
wall wart, 686
watt, 2
waveform
  sampling, 419
Wheeler's formula, 28
wideband
  analog link, 380
Widlar, Bob, 97, 101ff, 307, 598ff, 604, 679
Wien bridge, *see* oscillator
WiFi, 1062
Williams, Jim, 316, 330, 438, 1074, 1154
Willison, John, 968
Wilson mirror, *102*, *146*
window
  averaging, 562, 941g
  discriminator, *see* AoE 2nd ed., p 669
  quartz, 769, 1023
  sampling, 420–421
  switch, 573
wire
  long
    digital signals through, **856–874**, **1116–1122**
wired-AND, -OR, *see* logic
wirewound resistor, *see* resistor, wirewound
world's best
  DMM, 344
  low e_n amplifier, 505
  low distortion oscillator, 438
  low-memory capacitor (Teflon), 301g
  low-noise power supply, 580
  multichannel fast ΔΣ ADC, *956*
world's best (cont.)
  op-amp noise plot, 531g
  precision current source, *898*
  precision resistor, 227, 300, 1106
  PWM DAC, 889
  TIA, *547*
  voltage reference, 675, 677t, 681–682, 684, 892

X
XOR, *see* logic, gates

Y
yesterday, 273 (box)

Z
zener, *see* diode, zener
  active, 692
zero-crossing detector, *269*
ZigBee, 1061
ZVS, ZVC, *see* switchmode